

SUOMUSSALMI, Finland

WMO: 028790

Lat: 64.900N

Lon: 29.017E

Elev: 224

StdP: 98.66

Time Zone: 2.00 (EUE)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-25.8	-31.3	0.2	-28.8	-28.1	0.3	-25.7	8.3	-2.6	7.3	-2.9	0.3	150	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	23.8	16.0	22.1	15.5	20.4	14.7	17.5	21.6	16.4	20.3	15.5	19.2	2.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.6	19.6	14.7	10.8	18.0	13.7	10.1	17.1	50.3	21.7	46.9	20.4	44.1	19.3	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.4	-32.5	26.0	2.7	1.5	-34.4	27.1	-36.0	28.0	-37.5	28.8	-39.4	30.0		
			DB	WB	-32.6	18.8	2.6	1.0	-34.4	19.5	-36.0	20.1	-37.4	20.7	-39.3	21.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-10.4	-9.9	-5.5	-0.6	5.3	12.1	14.3	12.2	6.7	1.0	-6.0	-8.5
	DBStd	10.15	6.90	7.68	4.81	4.98	4.30	4.06	2.92	3.12	3.51	5.18	5.90	7.05
	HDD10.0	3635	632	558	482	318	160	23	3	14	110	281	479	574
	HDD18.3	6356	891	791	741	568	406	192	129	191	348	538	729	833
	CDD10.0	328	0	0	0	0	13	86	136	82	12	0	0	0
	CDD18.3	7	0	0	0	0	0	4	3	1	0	0	0	0
	CDH23.3	70	0	0	0	0	3	43	16	8	0	0	0	0
	CDH26.7	5	0	0	0	0	0	4	1	0	0	0	0	0
Wind	WSAvg	2.2	2.1	2.1	2.5	2.3	2.3	2.3	2.3	2.1	2.2	2.4	2.2	2.1
Precipitation	PrecAvg	635	44	36	35	31	53	61	82	73	63	60	55	50
	PrecMax	774	70	65	52	57	102	141	116	151	112	136	112	95
	PrecMin	461	10	4	8	11	11	27	24	26	8	8	9	10
	PrecStd	78	16	19	13	14	24	25	23	34	26	31	27	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.9	5.3	13.6	23.3	27.2	25.3	24.8	19.6	11.6	5.6	2.6
		MCWB	1.5	1.7	2.7	7.5	13.4	17.2	17.2	17.0	14.8	10.4	5.1	1.4
	2%	DB	1.0	2.2	3.4	10.9	19.8	24.5	23.1	22.4	16.0	9.9	3.7	1.6
		MCWB	0.3	1.3	1.0	6.0	11.9	15.7	16.2	15.8	12.0	8.9	3.1	1.0
	5%	DB	-0.2	0.7	1.9	8.8	16.4	22.0	21.7	20.5	13.7	9.0	2.5	0.8
		MCWB	-0.6	0.1	0.1	4.7	10.1	15.0	15.7	15.3	11.1	8.2	2.1	0.3
	10%	DB	-1.4	-0.3	0.7	6.7	13.3	19.8	20.2	18.5	12.1	7.8	1.1	0.0
		MCWB	-1.9	-0.8	-0.5	3.4	8.4	14.1	15.0	14.3	10.4	7.0	0.6	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.5	3.2	8.2	14.8	19.2	19.5	17.6	15.2	10.6	5.3	1.9
		MCDB	2.7	3.3	5.2	12.1	19.0	24.4	23.3	22.5	18.9	11.2	5.6	2.1
	2%	WB	0.5	1.3	1.3	6.5	12.5	17.5	17.7	16.8	13.0	9.2	3.2	1.1
		MCDB	1.0	1.8	2.6	10.1	19.4	21.3	21.4	21.2	15.1	9.8	3.5	1.4
	5%	WB	-0.6	0.2	0.5	5.1	10.5	16.0	16.5	15.6	11.8	8.3	2.1	0.4
		MCDB	-0.2	0.7	1.6	8.3	15.9	20.6	20.0	19.4	13.2	8.9	2.5	0.7
	10%	WB	-1.9	-0.8	-0.3	3.8	8.7	14.5	15.6	14.5	10.4	7.1	0.7	-0.5
		MCDB	-1.4	-0.3	0.6	6.1	12.7	19.1	19.0	18.0	11.9	7.7	1.1	-0.1
Mean Daily Temperature Range	5% DB	MDBR	7.1	6.6	7.7	8.5	8.8	8.7	8.5	8.4	6.6	4.4	5.2	5.8
		MCDBR	6.2	4.1	8.1	9.9	14.5	12.5	11.7	12.6	9.1	4.5	3.9	4.9
		MCWBR	6.1	3.9	6.4	6.1	7.7	6.1	6.2	7.0	5.7	4.0	4.0	4.9
	5% WB	MCDBR	6.3	4.1	6.2	9.3	13.4	9.9	9.4	11.0	7.8	4.6	4.0	4.6
		MCWBR	6.3	4.0	5.3	5.8	7.4	5.9	5.3	6.5	5.4	4.2	4.1	4.7
Clear-Sky Solar Irradiance	taub	0.199	0.269	0.298	0.332	0.339	0.339	0.361	0.341	0.317	0.294	0.350	0.198	
	taud	1.933	2.113	2.163	2.258	2.463	2.488	2.444	2.482	2.533	2.387	2.657	1.957	
	Ebn at Noon	316	615	771	835	871	877	842	828	774	607	300	91	
	Edh at Noon	24	59	89	104	94	94	95	83	64	46	21	8	
All-Sky Solar Radiation	RadAvg	0.12	0.63	2.05	3.75	4.65	5.14	4.82	3.57	1.95	0.69	0.16	0.04	
	RadStd	0.01	0.13	0.19	0.38	0.48	0.36	0.49	0.35	0.21	0.10	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page